

Zomato Restaurant Analytics

Exploratory Data Analysis (EDA) Using Python

Project Information

Project	Zomato Restaurant Analytics
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Role	Data Analyst
Project Type	Exploratory Data Analysis (EDA)
Domain	Restaurant & Food Delivery Analytics
Dataset	Zomato Restaurant Dataset

Tools & Technologies

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook

Project Objective

The objective of this project is to perform **Exploratory Data Analysis (EDA)** on the Zomato Restaurant Dataset to uncover meaningful insights into customer preferences, restaurant categories, pricing patterns, ratings, online ordering trends, and customer engagement.

The analysis aims to answer key business questions and provide actionable recommendations that can help restaurant owners and food delivery platforms make informed, data-driven decisions.

Project Overview

The objective of this project is to perform Exploratory Data Analysis (EDA) on the Zomato Restaurant dataset to uncover customer preferences, restaurant trends, and business insights.

The analysis focuses on:

- Restaurant categories
- Customer voting behaviour
- Ratings
- Online ordering
- Table booking
- Cost for two people

The project demonstrates data cleaning, preprocessing, visualization, and insight generation using Python.

Business Problem

Restaurant owners need to understand customer preferences to improve their services and increase customer satisfaction.

Using historical restaurant data, we aim to answer questions such as:

- what type of restaurant do the majority of customers order from ?
- How many votes has each type of restaurant received from customers ?
- What are the ratings that the majority of restaurants have received ?
- Zomato has observed that most couples order most of their food online. what is their average spending on each order ?
- Which mode (online or offline) has received the maximum rating ?
- Which type of restaurant received more offline orders , so that zomato can attract customers with good offers ?

Dataset Information

The dataset contains information about restaurants listed on the Zomato platform. It includes details related to restaurant services, customer ratings, voting patterns, pricing, and restaurant categories.

- **Number of Records:** 148

- **Number of Columns:** 7

Features

Column Name	Description
name	Name of the restaurant
online_order	Availability of online ordering (Yes/No)
book_table	Availability of table booking (Yes/No)
rate	Average customer rating
votes	Number of customer votes received
approx_cost(for two people)	Approximate cost for two people
listed_in(type)	Type or category of the restaurant

Import Required Librarie and Load Data set

```
In [5]: import pandas as pd
```

```
In [24]: df = pd.read_csv("Zomato data .csv")
```

In [25]: df

Out[25]:

	name	online_order	book_table	rate	votes	approx_cost two pec
0	Jalsa	Yes	Yes	4.1/5	775	800
1	Spice Elephant	Yes	No	4.1/5	787	800
2	San Churro Cafe	Yes	No	3.8/5	918	800
3	Addhuri Udupi Bhojana	No	No	3.7/5	88	300
4	Grand Village	No	No	3.8/5	166	600
...	...	...	...	...	...	...
143	Melting Melodies	No	No	3.3/5	0	100
144	New Indraprasta	No	No	3.3/5	0	150
145	Anna Kuteera	Yes	No	4.0/5	771	450
146	Darbar	No	No	3.0/5	98	800
147	Vijayalakshmi	Yes	No	3.9/5	47	200

148 rows × 7 columns



In [26]: df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 148 entries, 0 to 147
Data columns (total 7 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   name                                  148 non-null    object
1   online_order                         148 non-null    object
2   book_table                           148 non-null    object
3   rate                                 148 non-null    object
4   votes                                148 non-null    int64
5   approx_cost(for two people)          148 non-null    int64
6   listed_in(type)                       148 non-null    object
dtypes: int64(2), object(5)
memory usage: 8.2+ KB
```

```
In [27]: import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

we should convert the data type of rate column

```
In [29]: df['rate'] = df['rate'].apply(lambda x: str(x).split('/')[0] if '/' in
str(x) else x)
```

```
In [11]: df
```

Out[11]:

	name	online_order	book_table	rate	votes	approx_cost two people
0	Jalsa	Yes	Yes	4.1	775	800
1	Spice Elephant	Yes	No	4.1	787	800
2	San Churro Cafe	Yes	No	3.8	918	800
3	Addhuri Udupi Bhojana	No	No	3.7	88	300
4	Grand Village	No	No	3.8	166	600
...	...	...	...	...	...	...
143	Melting Melodies	No	No	3.3	0	100
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146	Darbar	No	No	3.0	98	800
147	Vijayalakshmi	Yes	No	3.9	47	200

148 rows × 7 columns



```
In [12]: df['rate'] = df['rate'].astype(float)
```

In [13]: `df.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 148 entries, 0 to 147
Data columns (total 7 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   name                                  148 non-null    object
1   online_order                         148 non-null    object
2   book_table                           148 non-null    object
3   rate                                 148 non-null    float64
4   votes                                148 non-null    int64
5   approx_cost(for two people)         148 non-null    int64
6   listed_in(type)                      148 non-null    object
dtypes: float64(1), int64(2), object(4)
memory usage: 8.2+ KB
```

Problem statements

1. what type of restaraunt do the majority of customers order from ?

In [14]: `df.head()`

Out[14]:

	name	online_order	book_table	rate	votes	approx_cost(for two people)
0	Jalsa	Yes	Yes	4.1	775	800
1	Spice Elephant	Yes	No	4.1	787	800
2	San Churro Cafe	Yes	No	3.8	918	800
3	Addhuri Udupi Bhojana	No	No	3.7	88	300
4	Grand Village	No	No	3.8	166	600

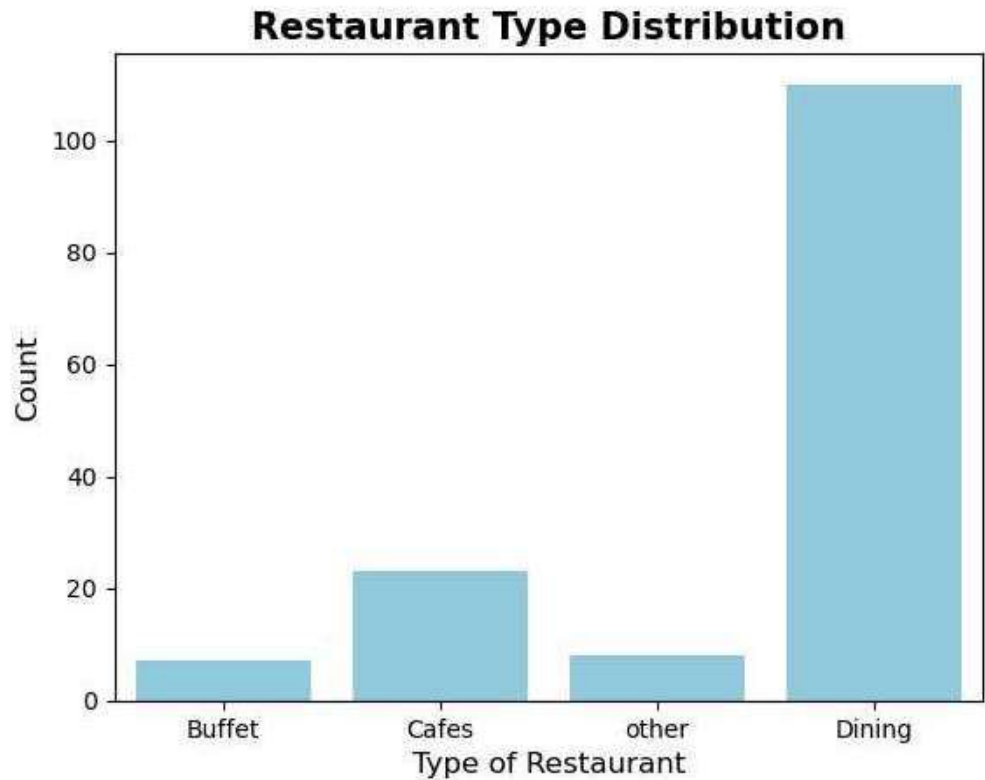
In [15]: `df['listed_in(type)'].unique()`

Out[15]: array(['Buffet', 'Cafes', 'other', 'Dining'], dtype=object)

```
In [38]: sns.countplot(x='listed_in(type)', data=df, color='skyblue')

plt.xlabel('Type of Restaurant', fontsize=12)
plt.ylabel('Count', fontsize=12)
plt.title('Restaurant Type Distribution', fontsize=15, fontweight='bold')

plt.show()
```



Conclusion -the majority of customers order from Dining category restaurant

2. How many votes has each type of restaurant received from customers ?

```
In [17]: df.head(2)
```

Out[17]:

	name	online_order	book_table	rate	votes	approx_cost(for two people)
0	Jalsa	Yes	Yes	4.1	775	800
1	Spice Elephant	Yes	No	4.1	787	800